

Lab Task 3: Bank Account Management System with Inheritance and Polymorphism

Objective:

In this task, you will extend the concepts from Lab Task 2, which focused on creating a basic bank account system. The goal now is to enhance the system by adding multiple types of bank accounts (e.g., Savings, Checking) using Inheritance and Polymorphism. This will give you a deeper understanding of object-oriented programming (OOP) concepts in C#, such as class inheritance, method overriding, and polymorphic behavior.

Instructions:

1. Extend the BankAccount Class:

- You should use the BankAccount class from **Lab 2** as the base class.
- Create two new derived classes: SavingsAccount and CheckingAccount.

2. Derived Classes:

- **SavingsAccount Class:**
 - **Instance Properties:**
 1. Interest Rate = 4% [Const]
 - This class should inherit from BankAccount.
 - It should override the DisplayAccountInfo() method to show additional information, like the interest rate.
 - It should have a method to apply interest to the balance (ApplyInterest()). Use base.Desposit() to deposit interest amount

3. Test the Bank Accounts:

- Create instances of SavingsAccount in your Main method and perform operations like deposits, withdrawals, ApplyInterest, and displaying account information.
- Use a **base class reference** to call overridden methods of the derived classes.